

AC-134

A/C Service Station

BDB Mechanixpro Tools

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Safety

- The machine is designed to be used and repaired by qualified personnel only.
- The machine designed for use in recovering the R134a refrigerant fluid used in the air-conditioning (A/C) systems of motor vehicles. Fill the A/C system with the quantity of refrigerant recommended by the manufacturer.
- Check the vehicle use and maintenance manual for the type of refrigerant fluid used by the A/C system. Do not mix different type of refrigerant otherwise can easily lead to malfunction of the machine.
- Keep away from moving parts and rotating elements such as cooling fans, alternators and heating components, etc to avoid harm.
- Wear protective clothing gloves and goggles.
- Do not expose the machine to direct sunlight or rain. Use only in well-ventilated work areas.
- Never exceed 30 ° tilt in transit process upside down is strictly prohibited.
- Do not touch the machine high voltage power supply section, and do not maintain the machine as power on.
- Care of the manual.
- We reserve the right to modify the contents of this document without prior notice to our Customers

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1. Introduction

1) Outline

A/C system maintenance equipment is with the latest design technique which use the best control principle and the manufacturing process.

A/C Service Station is intelligent equipment collecting of the recovery, recycling, recharging and other functions in one. It's with the beautiful shape, humanized operation interface, and advanced manufacturing processes to make the A/C maintenance professional and simple.

2) Features

- A. All functions running are controlled by solenoid valves.
- B. Unique design of the pipeline to achieve the high recycling rate no matter with the gas or the liquid.
- C. Easy to operate with the concise operating interface.

3) Specifications

- A. Voltage input: AC110V $\pm$ 10%~50/60Hz
- B. Compressor: 12.12cm³
- C. Vacuum pump: 7.2m³/h, 5Pa
- D. Load cell: $\pm$ 10g
- E. Tank: 12L
- F. Oil bottle: 260ml,double
- G. Working pressure: max. 20bar
- H. HP gauge: -1bar~3.5MPa
- I. LP gauge: -1bar~1.5MPa
- J. Tank pressure gauge: -1bar~3.5MPa

2. Functions

1) Primary functions

- A. Recovery/Recycling
- B. Auto./Manual used oil discharging
- C. Vacuum
- D. New oil filling
- E. Recharging

2) Subsidiary functions

- A. Refrigerant supplying
- B. Load cell calibration
- C. Parts checking
- D. Maintenance procedure for dry filter
- E. Alarms
- F. Parameters setting
- G. Parameters inquiring

3. Operation

1) Parts descriptions





NOTE:  This key is backup and no useful under this mode.

2) First use

A、 Unscrew the electronic scale support screw before using the equipment.


Unscrew the
support screw.

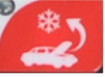


From the back.

- B、 When power on machine, the LED will displays "F001" and can clear it by pressing "STOP" key. Please supply about 4 kg refrigerants into the tank of equipment so that all of the functions can be run normally. Please refer to the chapter of **"Refrigerant supply"**.
- C、 Please read this user manual carefully before operating.

3) Recovery/Recycling

- A、 Start the machine then press "" key to select recovery function.
 - B、 Presses "Start" key to run.
 - C、 It will stop running and discharge the used oil automatically when the recovery/recycling process finish.
- Note: also be able to discharge the used oil by press the switch.**
- D、 During the running process, if it displays the code F002 or F003, please refer to the APPENDIX: ALARM CODE TABLE.

Note: Press "" and "STOP" key in recovery function. It will purify the non-condensable gas automatically.

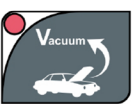
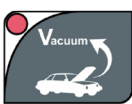
4) Vacuum

- A、 Press "" key.
- B、 Set the vacuum time by pressing "+" or "-" and then press the ENTER" key to confirm it. It should be more than 15 minutes. Generally, it needs 15 minutes at least for the air conditioning only with front wind and 20 minutes for with the front and rear wind.
- C、 Press "START" key to run it.
- D、 When the time is finished, it will stop automatically.
- E、 If it displays code F007 during the process, please refer to the APPENDIX: ALARM CODE TABLE.

5) New oil adding

- A、 Pour new oil into the new oil bottle.

Note: add new oil more 20ml than the used oil drained out to avoid the air into the air conditioning system.

- B、 Press , Select vacuum function。 Then press  and  key at the same time, The display now shows "OIL".

- C、 Press  to add new oil

Press the key once and fill with new oil once, about 5 ~ 10 ml (according to the vacuum degree of the system).

If you press and hold the key, fill new oil continuously at this time. After releasing the key, stop filling.

D、 If the equipment displays code "F011", it indicates that the air conditioning system is in a non vacuum state and cannot be filled with new oil. Refill after vacuumizing again.



6) Recharging



A、 Press the “ ” key.

B、 Set filling amount by press “+” or “-” key, and then press “ENTER” key.

Note: set the filling amount according to automotive user manual. The unit is kg.

C、 Press “START” key to run.

D、 Stop automatically after recharging finished.

Note: In the filling process, if LED displays alternately the remaining amount of filling and the letters "AC", it means that you need to turn on the car air-conditioning to complete the filling process. Later, you should supply new refrigerant into the tank.

7) Refrigerant supply

A、 Connect the port of fresh refrigerant tank to the HP port of equipment. Block the LP port of equipment with a cap or LP coupler.

Note: it should connect to the liquid port of fresh refrigerant tank. If has no liquid port, please invert the fresh refrigerant tank.

B、 Press and hold the "-" key and power on the equipment at same time. The “RECOVERY” and “Kg” indicators will light on, which means refrigerant supply function is on.

C、 Set supply amount by press “+” or “-” key, and then press “ENTER” key.

D、 Press “START” key to run.

E、 When the LCD display “CLO”, please close the valve of the fresh refrigerant tank.

F、 Stop automatically after resupply finished.

8) Usage of the Auxiliary Port

A、 Leak checking by N₂

- i. Remove the cap of auxiliary port after the equipment stop working entirely.
- ii. Connect the N₂ to the auxiliary port.

Note: the connector to the auxiliary port must be with thimble. Do not open the valve of N₂ during this process.

- iii. Adjust the N₂ pressure to a suitable value and open the valve of N₂. When the pressure is stable, please close the valve of N₂. Normally, the pressure is adjusted to about 12~15bar.
- iv. Check if there has any leakage in the A/C system. After checking, please press the needle in the auxiliary port to relive the N₂.
- v. After finished, re-back the cap to the auxiliary port.

B、Recharging from external tank

- i. Before vacuum, connect the pipe with the external tank to the auxiliary port.

Note: the connector to the auxiliary port must be with thimble. Do not open the valve of external tank during this process.

- ii. Run vacuum function.
- iii. Open the valve of external tank to recharge the right amount refrigerant into A/C system. And then close the valve of external tank.
- iv. After finished, disconnect the pipes and re-back the cap to the auxiliary port.

9) Maintenance

A、Calibration of load cell

When the load cell is not precise, it needs to be calibrated.

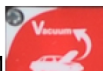
Calibrating procedure as below:

- i. Prepare a 15 kg weight. And remove the refrigerant tank from scale pan and keep the pan empty.
Note: do not disconnect the red and blue hose from the refrigerant tank. It's better put the refrigerant tank on a lower stage.
- ii. Press and hold "STOP" key then power on the equipment. The LED displays "1" in flushing.
- iii. Press "ENTER" key to do the empty plan calibration. Then the LED will display "2" in flushing.
Note: If LED shows "ERR1", please refer to the APPENDIX: ALARM CODE TABLE.
- iv. Put the 15 kg weight onto the scale pan and press "ENTER" key. Then LED displays the weight value and adjusts the value to 15 by press "+" or "-"key.
Note: if the weight is 20kg, the value must be set to 20.
- v. Press "ENTER" key to do the scale linear calibration. Then the LED displays "END" to mean it's successful for the calibration.
Note: if LED shows "ERR2", please refer to the APPENDIX: ALARM CODE TABLE.
- vi. Then power off and replace and fasten the refrigerant tank onto the scale pan.

B、Exchange dry filter

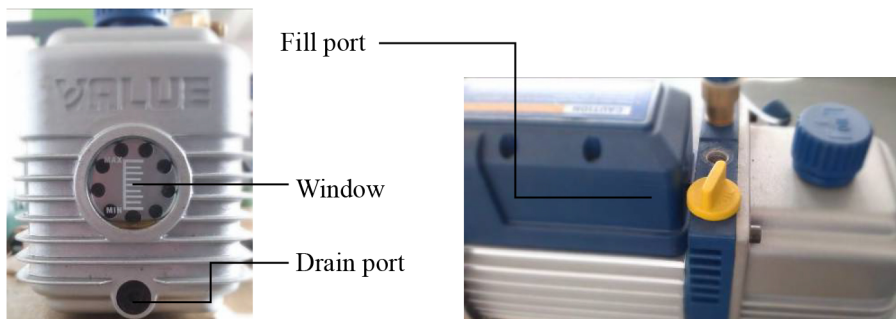
The dry filter must be exchange after used a period. The machine has set the exchange time.

When the dry filter reaches the life time, the LED will show code "F004". Now the dry filter must be exchanged.



- i. Press and hold “VACUUM” key and power on the equipment.
- ii. Press “START” key to do the first step automatically.
- iii. Replace the used dry filter with a new one then press “ENTER” key.
- iv. Input the dry filter ID. Please refer to the chapter **PARAMETER SETTING**.
- v. Power off and dry filter exchange procedure finish.

C、Change vacuum pump oil



- i. When the pump oil became cream or the LED displays “F008”, the vacuum pump oil must be changed.
- ii. Step 1: unscrew the block of the drain port to drain entirely out the old oil. Then re-back the block to the drain port.
- iii. Step 2: unscrew the cap of the fill port and then fill the new oil slowly into the vacuum pump until the oil level reach at the center site. Then re-back the cap to the fill port.

Note: the new vacuum pump oil cannot be filled too much into the vacuum pump otherwise it will spray out when working.

10) Parameter Inquiring



- i. Press and hold “Recharging” key to enter parameter inquiring mode.
- ii. In this mode three parameters can be inquired.
The LED displays the parameter code and you can change the code by press “+” or “-” key.
- iii. Code “C”: total worked time of the compressor
When LED displays code "C", please press “ENTER” key to show the total worked time.
- iv. Code “P”: total worked time of the vacuum pump
When LED displays code "P", please press “ENTER” key to show the total worked time.
- v. Code “F”: volume of refrigerant dealt with by dry filter.
When LED displays code "P", please press “ENTER” key to show the total refrigerant amount.

Reset this code by press together “Fill” and “STOP” keys and the ways of setting dry-filter ID is as follows:

Dry filter ID is with 4 digits, and you need to set one by one.

When one digit is in flash, it means you can set it by press “+” or “-” key, and press “ENTER” key to confirm it.

Set a same value according to the ID pasted on the dry-filter.

Only you set the correct 4 digits, the equipment can accept it

Note: The code used already will be not able to be accepted again.

APPENDIX: ALARM CODE TABLE

Code	Descriptions	Emergency Method
F001	No enough refrigerant in cleaning tank	Supply refrigerant in time.
F002	Too much refrigerant in tank	Transfer some refrigerant to another tank.
F003	High pressure	Stop use for a period.
F004	No refrigerant to recovery	Run next function.
F005	Cannot vacuum	Recovery first and then run vacuum.
F006	Exchange dry-filter	The dry-filter is life-end.
F007	Vacuum pump maintenance	Change the pump oil in time.
F009	No enough refrigerant in recharging tank	Change a new tank of refrigerant.
F010	Prohibit changing the recharging tank	Here has enough refrigerant in the tank.
ERR1	Load cell calibration error at step 1	Check if the scale pan empty. Check the connection of signal cable being ok. Check if the load cell is damaged.
ERR2	Load cell calibration error at step 2	Check if the weight has been put onto the scale pan. Check the connection of signal cable being ok. Check if the load cell is damaged.

Declare:

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